The details of this Test Plan have been extracted from the Research Requirements Document for Droplet Flame Extinguishment in Microgravity (FLEX) by FA Williams, M Hicks, V Nayagam, MY Choi, FL Dryer and BD Shaw and dated November 4, 2005.

2.2 Agent Efficacy

The expressions for the Damköhler number provide a convenient theoretical basis for flame extinction. Unfortunately, they are not easily used by engineers designing suppression systems or evaluating material flammability issues. For a given material and ambient condition (pressure, diluent type), there exists an oxygen mole fraction below which a steady flame cannot exist. This oxygen concentration is called the Limiting Oxygen Index (LOI). Using the droplet as a model geometry, Figure 2 shows a hypothetical quasi-steady flammability map. The graph shows oxygen concentration on the ordinate and droplet diameter on the abscissa. The curve is essentially the critical Damköhler number and the region inside the curve is flammable and outside the region is non-flammable. The red branch of the curve is the extinction branch defined by Law (1975) and the blue branch is the radiative quench branch defined by Chao et al. (1990). A typical droplet combustion experiment starts by igniting a droplet at a point inside the curve (denoted by the green circle). The droplet then burns until it reaches the red curve where the flame will extinguish. The limiting oxygen index (LOI) is the point where the two curves meet, or the minimum oxygen concentration that will support a quasi-steady flame.

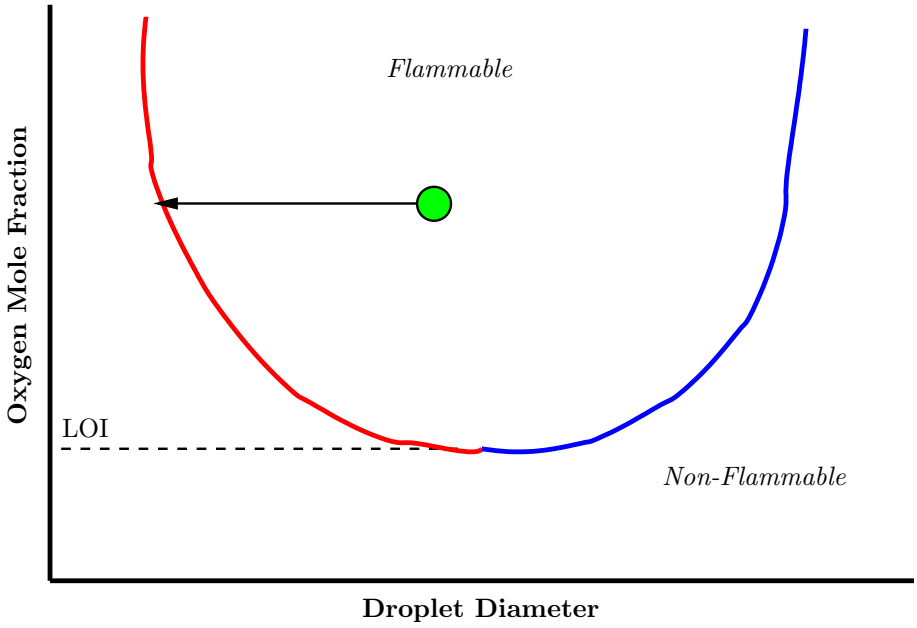


Figure 2: A conceptual flammability map for a single droplet burning in an infinite ambient. The red curve represents the extinction caused by 'blowoff' of the flame (Law, 1975), while the blue curve represents the radiative extinction branch (Chao et al., 1990). The LOI is the lowest oxygen concentration which will support combustion.

The LOI is a critical concept in fire safety, and also important in developing, refining and ultimately validating detailed chemical kinetic mechanisms. For fire safety, the goal of the agent in a total flooding application is to reduce the oxygen concentration below the LOI .

The concepts of extinction and LOI reviewed above define the framework from which to develop a quantitative measure of suppressant agent efficacy. The LOI is often recast in terms of the Minimum Extinction Concentration (the minimum concentration of a suppressant agent above which a quasi-steady flame cannot exist, MEC). The MEC is related to the LOI by the following relationship.

Table 2: Gaseous fire suppression agent efficacy, as measured by the MEC , for cup burner flames (Takahashi et al., 2004).

Agent/Env.	MEC , $1-g$	MEC , μg
CO_2	0.16	0.19
He	0.27	0.31
N_2	0.26	0.38

$$LOI = 0.209\,(1 - MEC) \tag{2}$$

The MEC is a function of the fuel, agent and experimental geometry. Clearly, a more effective agent will have a lower MEC for a given fuel and fuel geometry.

Takahashi et al. (2004) conducted normal gravity tests and microgravity simulations of the suppression of methane cup-burner flames using mixtures of various suppressants in standard air. The table below summarizes the findings for MEC in the two environments.

2.3 CO_2 Suppression Studies

The United States module of the ISS uses CO_2 as the suppressant. The effectiveness of CO_2 as a fire suppressant in a microgravity environment has been studied in the past by several authors (Honda and Ronney, 1998; Katta et al., 2004). Honda and Ronney (1998) studied opposed-flow flame spread rates over thin solid fuels in microgravity by adding different inert diluents (He , N_2 , Ar, CO_2 and SF_6) to the standard air atmosphere. Their experimental results seem to suggest that CO_2 is less effective as a fire suppressant in microgravity than He in terms of the minimum O_2 concentration that supports combustion. Katta et al. (2004) investigated the mechanism through which CO_2 causes fire suppression using detailed numerical simulation and experiments in microgravity for a methane-air diffusion flame stabilized on a cup burner. Their results showed that a CO_2 volume fraction greater than 19.1% is needed to destabilize the flame-base in microgravity compared to 16.4% for a normal gravity flame. They also pointed out the importance of gas-phase radiation in determining extinguishment limits in microgravity.

Radiative thermal losses from the flame zone in reduced gravity environments may be significantly altered when the surrounding gas phase is thermally participating. In most small scale terrestrial fires, when compared with the quenching effects associated with strong buoyantly driven convective flows, radiative quenching plays a relatively minor role in the extinguishment of fires. In these situations the heated gases are quickly lifted from the reaction zone and, as a consequence, quickly cool and provide little radiative feedback to the fuel source.

In the absence of buoyantly-driven convective flows, however, radiative losses from the flame can become a significant fraction of the total heat loss from the reaction zone. When radiative exchanges between the flame and the ambient are limited by the presence of a thermally participating gas, such as CO_2 , temperatures in the proximity of the fuel source will increase. Chernovsky et al. (2005) demonstrated this in a recent set of experiments where fuel side temperature increases were reported in an ethylene (C_2H_2) diffusion flame, using a porous spherical burner, in atmospheres of varying concentrations of CO_2 (20% to 40%).

This radiative feedback effect from the gas phase has been of increasing concern because of the implications in the selection and evaluation of appropriate fire suppressants. The use of CO_2 , an

5.2.8 Ambient Temperature and Pressure Measurement

During each test the chamber gas temperature and pressure measurements shall be required. The following are the specifications for this data.
Sample rate: minimum sample rate shall be at 10 Hz
Temperature: accuracy shall be at least $\pm\, 0.5C$.
Pressure: accuracy shall be at least $\pm\, 0.01\ atm$.

5.2.9 Synchronization

Experiment objectives require all data to be time-synchronized to a reference time. This will allow accurate interpretation of the data and evaluate droplet regression rate, flame structure, and flame extinction as a function of time.
Time synchronization: all measurements shall be referenced to GMT with a minimum accuracy of $\pm\, 0.03\ sec$.

5.3 Flight Experiment Test Procedures

The FLEX experiments involve igniting and observing the burning behavior of relatively large droplets. One of the primary objectives of the experiments is to determine the LOI as a function of ambient pressure and determine how the presence of a suppressant influences the LOI . The determination of the LOI involves conducting a series of tests and constructing a diagram similar to that in Fig. 2. The basic outline of the procedure to construct this diagram is as follows:

1. Select the test pressure ($0.5\ -\ 1.5\ atm$) for the LOI tests.
2. Select a smaller droplet size ($\approx\ 3\ mm$)
3. Select an ambient oxygen mole fraction that, based on ground-based testing, will support droplets burning to completion (as opposed to extinguishing at a finite droplet size).
4. Perform a droplet combustion test.
5. Increase the droplet size or repeat the test (if finite extinction diameter at flame extinction observed).
6. Continue steps 4 and 5 until flame exists for only a short time after ignition (radiative extinction).
7. Continue steps 4 thru 6 at a decreased ambient oxygen mole fraction.
8. Continue step 7 to an ambient oxygen mole fraction where the droplet does not ignite.

There will likely be ambient conditions where the extinction droplet sizes at all ambient oxygen mole fractions are too small to either be observed or measured with the available diagnostics. In this case, the determination of the LOI is similar, but more iterative. Instead of observing flame extinction, however, the procedure will involve an iterative sequence of tests (burn to completion followed by a no ignition test) with progressively smaller increments in ambient oxygen mole fraction to determine the LOI .

For the slow flow tests (in the flight experiment), the current configuration of the flight hardware does not allow for adequate diagnostics over the entire test for a droplet in a uniform, steady

convective flow[7]. In these tests, the procedure will be to duplicate tests above which yield finite-sized extinction droplet diameters. Near the end of the tests, prior to flame extinction, the droplet will by subjected to a small, sub-buoyant, convective flow (imparted by translating the droplet at a slow, uniform speed). The goal of the experiment is then to see how the presence of this flow influences the extinction process. We also note that a large number of the tests will involve free droplets (as opposed to fiber-supported droplets). A number of these tests will have droplets with a slow drift velocity. We can also use the data gathered from these tests to examine the influences of a small, sub-buoyant flow on both the burning and extinction process.

The procedure to examine the suppressant efficacy will be essentially the to that used to determine the LOI . The major difference will be that in addition to the diluent (N_2), the ambient will contain a suppressant. The test procedure will be identical to that above; the steps described above will be performed at a fixed suppressant level. Subsequent series of tests will then be at different suppressant concentrations and types of suppressants.

5.4 FLEX Test Matrix

The proposed test matrix is fashioned so as to meet the stated science objectives and at the same time meet the constraints imposed by the hardware as well as the safety requirements of space experiments. The previous section described the test procedures. The exact test matrix will change during the conduct of the FLEX experiment. The initial conditions for the first series of FLEX tests will be determined by extensive ground testing in the drop towers at the NASA Glenn Research Center. It is important, however, for the engineering team to have a test matrix on paper so that it can determine consumable resource requirements and prepare for the necessary safety reviews. The following test matrices, then are designed to allow the engineering team to adequately plan for the experiment and do not represent the final test matrix. These test matrices provide for an envelope in which all of the actual tests will be conducted.

The test matrix is broken up into a series of sub-matrices each of which is designed to highlight an objective of the experiment. The first series of tests, described in Tables 6 and 7, will determine the LOI as a function of pressure for heptane and methanol in quiescent and flowing environments, respectively. The second series of tests, described in Tables 8, 9 and 10, examines the efficacy of CO_2 , He and SF_6 . The tests use the two test fuels, are over a range of pressures, oxygen mole fractions and suppressant mole fractions. The tests described in Table 11 examine how a slow, sub-buoyant flow influences the efficacy of a single suppressant, CO_2 . Finally, the tests in Table 12 examine the influence of a stronger radiative environment (achieved by increasing the pressure) on the efficacy of a single suppressant.

[7]This will, however, be extensively studied in the complimentary ground-based program

Table 6: Quiescent environment LOI test matrix.

Fuel: C_7H_{16} , Diluent: N_2 , No fiber support, Flow: $0.0\,cm/s$, $D_0 = 3,\,5\,mm$, $X_{N2} = 1.0 - X_{O2}$.		
Test No.	$P\,(atm)$	X_{O2}
1	0.50	0.20
2	0.50	0.30
3	0.50	0.35
4	0.75	0.13
5	0.75	0.20
6	0.75	0.28
7	0.75	0.40
8	1.00	0.10
9	1.00	0.15
10	1.00	0.21
11	1.00	0.30
12	1.50	0.10
13	1.50	0.15
14	1.50	0.21
Fuel: CH_3OH		
15	0.50	0.20
16	0.50	0.30
17	0.50	0.35
18	0.75	0.13
19	0.75	0.20
20	0.75	0.28
21	0.75	0.40
22	1.00	0.10
23	1.00	0.15
24	1.00	0.21
25	1.00	0.30
26	1.50	0.10
27	1.50	0.15
28	1.50	0.21

Table 7: Slow flow LOI test matrix

Fuel: C_7H_{16} , Diluent: N_2 , SiC fiber support, Flow: $1.5\,cm/s$, $D_0 = 3,\ 5\,mm$, $X_{N2} = 1.0 - X_{O2}$.		
Test No.	$P\,(atm)$	X_{O2}
29	0.50	0.20
30	0.50	0.30
31	0.50	0.35
32	0.75	0.13
33	0.75	0.20
34	0.75	0.28
35	0.75	0.40
36	1.00	0.10
37	1.00	0.15
38	1.00	0.21
39	1.00	0.30
Fuel: CH_3OH		
40	0.50	0.20
41	0.50	0.30
42	0.50	0.35
43	0.75	0.13
44	0.75	0.20
45	0.75	0.28
46	0.75	0.40
47	1.00	0.10
48	1.00	0.15
49	1.00	0.21
50	1.00	0.30

Table 8: CO_2 suppressant efficacy test matrix.

Fuel: C_7H_{16} , Diluent: N_2 , Suppressant: CO_2 , No fiber support, Flow: $0.0\,cm/s$, $D_0 = 3,\ 5\,mm$, $X_{N2} = 1.0 - X_{O2} - X_{CO2}$.			
Test No.	$P\,(atm)$	X_{O2}	X_{CO2}
51	0.70	0.25	0.15
52	0.70	0.25	0.25
53	0.70	0.25	0.50
54	0.70	0.25	0.70
55	0.70	0.30	0.15
56	0.70	0.30	0.25
57	0.70	0.30	0.50
58	0.70	0.30	0.70
59	0.85	0.25	0.15
60	0.85	0.25	0.25
61	0.85	0.25	0.50
62	0.85	0.25	0.70
63	0.85	0.30	0.15
64	0.85	0.30	0.25
65	0.85	0.30	0.50
66	0.85	0.30	0.70
67	1.00	0.10	0.15
68	1.00	0.10	0.25
69	1.00	0.10	0.50
70	1.00	0.10	0.70
71	1.00	0.15	0.15
72	1.00	0.15	0.25
73	1.00	0.15	0.50
74	1.00	0.15	0.70
75	1.00	0.21	0.15
76	1.00	0.21	0.25
77	1.00	0.21	0.50
78	1.00	0.21	0.70
Fuel: CH_3OH .			
79	0.70	0.25	0.15
80	0.70	0.25	0.25
81	0.70	0.25	0.50
82	0.70	0.25	0.70
83	0.70	0.30	0.15
84	0.70	0.30	0.25
85	0.70	0.30	0.50
86	0.70	0.30	0.70
87	0.85	0.25	0.15
88	0.85	0.25	0.25
89	0.85	0.25	0.50
90	0.85	0.25	0.70

91	0.85	0.30	0.15
92	0.85	0.30	0.25
93	0.85	0.30	0.50
94	0.85	0.30	0.70
95	1.00	0.10	0.15
96	1.00	0.10	0.25
97	1.00	0.10	0.50
98	1.00	0.10	0.70
99	1.00	0.15	0.15
100	1.00	0.15	0.25
101	1.00	0.15	0.50
102	1.00	0.15	0.70
103	1.00	0.21	0.15
104	1.00	0.21	0.25
105	1.00	0.21	0.50
106	1.00	0.21	0.70

Table 9: He suppressant efficacy test matrix.

Fuel: C_7H_{16} , Diluent: N_2 , Suppressant: He , No fiber support, Flow: $0.0\,cm/s$, $D_0 = 3,\ 5\,mm$, $X_{N2} = 1.0 - X_{O2} - X_{He}$.			
Test No.	$P\,(atm)$	X_{O2}	X_{He}
107	0.70	0.25	0.15
108	0.70	0.25	0.25
109	0.70	0.25	0.50
110	0.70	0.25	0.70
111	0.70	0.30	0.15
112	0.70	0.30	0.25
113	0.70	0.30	0.50
114	0.70	0.30	0.70
115	0.85	0.25	0.15
116	0.85	0.25	0.25
117	0.85	0.25	0.50
118	0.85	0.25	0.70
119	0.85	0.30	0.15
120	0.85	0.30	0.25
121	0.85	0.30	0.50
122	0.85	0.30	0.70
123	1.00	0.10	0.15
124	1.00	0.10	0.25
125	1.00	0.10	0.50
126	1.00	0.10	0.70
127	1.00	0.15	0.15
128	1.00	0.15	0.25
129	1.00	0.15	0.50
130	1.00	0.15	0.70
131	1.00	0.21	0.15
132	1.00	0.21	0.25
133	1.00	0.21	0.50
134	1.00	0.21	0.70
Fuel: CH_3OH			
135	0.70	0.25	0.15
136	0.70	0.25	0.25
137	0.70	0.25	0.50
138	0.70	0.25	0.70
139	0.70	0.30	0.15
140	0.70	0.30	0.25
141	0.70	0.30	0.50
142	0.70	0.30	0.70
143	0.85	0.25	0.15
144	0.85	0.25	0.25
145	0.85	0.25	0.50
146	0.85	0.25	0.70

147	0.85	0.30	0.15
148	0.85	0.30	0.25
149	0.85	0.30	0.50
150	0.85	0.30	0.70
151	1.00	0.10	0.15
152	1.00	0.10	0.25
153	1.00	0.10	0.50
154	1.00	0.10	0.70
155	1.00	0.15	0.15
156	1.00	0.15	0.25
157	1.00	0.15	0.50
158	1.00	0.15	0.70
159	1.00	0.21	0.15
160	1.00	0.21	0.25
161	1.00	0.21	0.50
162	1.00	0.21	0.70

Table 10: SF_6 suppressant efficacy test matrix.

Fuel: C_7H_{16} , Diluent: N_2 , Suppressant: SF_6 , No fiber support, Flow: $0.0\,cm/s$, $D_0 = 3,\ 5\,mm$, $X_{N2} = 1.0 - X_{O2} - X_{SF6}$.			
Test No.	$P\,(atm)$	X_{O2}	X_{SF6}
163	0.70	0.25	0.15
164	0.70	0.25	0.25
165	0.70	0.25	0.50
166	0.70	0.25	0.70
167	0.70	0.30	0.15
168	0.70	0.30	0.25
169	0.70	0.30	0.50
170	0.70	0.30	0.70
171	0.85	0.25	0.15
172	0.85	0.25	0.25
173	0.85	0.25	0.50
174	0.85	0.25	0.70
175	0.85	0.30	0.15
176	0.85	0.30	0.25
177	0.85	0.30	0.50
178	0.85	0.30	0.70
179	1.00	0.10	0.15
180	1.00	0.10	0.25
181	1.00	0.10	0.50
182	1.00	0.10	0.70
183	1.00	0.15	0.15
184	1.00	0.15	0.25
185	1.00	0.15	0.50
186	1.00	0.15	0.70
187	1.00	0.21	0.15
188	1.00	0.21	0.25
189	1.00	0.21	0.50
190	1.00	0.21	0.70
Fuel: CH_3OH			
191	0.70	0.25	0.15
192	0.70	0.25	0.25
193	0.70	0.25	0.50
194	0.70	0.25	0.70
195	0.70	0.30	0.15
196	0.70	0.30	0.25
197	0.70	0.30	0.50
198	0.70	0.30	0.70
199	0.85	0.25	0.15
200	0.85	0.25	0.25
201	0.85	0.25	0.50
202	0.85	0.25	0.70

203	0.85	0.30	0.15
204	0.85	0.30	0.25
205	0.85	0.30	0.50
206	0.85	0.30	0.70
207	1.00	0.10	0.15
208	1.00	0.10	0.25
209	1.00	0.10	0.50
210	1.00	0.10	0.70
211	1.00	0.15	0.15
212	1.00	0.15	0.25
213	1.00	0.15	0.50
214	1.00	0.15	0.70
215	1.00	0.21	0.15
216	1.00	0.21	0.25
217	1.00	0.21	0.50
218	1.00	0.21	0.70

Table 11: Suppressant efficacy in a slow flow test matrix.

Fuel: C_7H_{16} , Diluent: N_2 , Suppressant: CO_2 , No fiber, Flow: $1.5\,cm/s$, $D_0 = 3,\ 5\,mm$, $X_{N2} = 1.0 - X_{O2} - X_{CO2}$.			
Test No.	$P\,(atm)$	X_{O2}	X_{CO2}
219	0.70	0.10	0.15
220	0.70	0.10	0.25
221	0.70	0.10	0.50
222	0.70	0.10	0.70
223	0.70	0.21	0.15
224	0.70	0.21	0.25
225	0.70	0.21	0.50
226	0.70	0.21	0.70
227	1.00	0.10	0.15
228	1.00	0.10	0.25
229	1.00	0.10	0.50
230	1.00	0.10	0.70
231	1.00	0.15	0.15
232	1.00	0.15	0.25
233	1.00	0.15	0.50
234	1.00	0.15	0.70
235	1.00	0.21	0.15
236	1.00	0.21	0.25
237	1.00	0.21	0.50
238	1.00	0.21	0.70
Fuel: CH_3OH			
239	0.70	0.10	0.15
240	0.70	0.10	0.25
241	0.70	0.10	0.50
242	0.70	0.10	0.70
243	0.70	0.21	0.15
244	0.70	0.21	0.25
245	0.70	0.21	0.50
246	0.70	0.21	0.70
247	1.00	0.10	0.15
248	1.00	0.10	0.25
249	1.00	0.10	0.50
250	1.00	0.10	0.70
251	1.00	0.15	0.15
252	1.00	0.15	0.25
253	1.00	0.15	0.50
254	1.00	0.15	0.70
255	1.00	0.21	0.15
256	1.00	0.21	0.25
257	1.00	0.21	0.50
258	1.00	0.21	0.70

Table 12: Suppressant in a radiative environment test matrix.

Fuel: C_7H_{16} , Diluent: N_2 , Suppressant: CO_2 , No fiber support, Flow: $0.0\,cm/s$, $D_0 = 3,\ 5\,mm$, $X_{N2} = 1.0 - X_{O2} - X_{CO2}$.			
Test No.	$P\,(atm)$	X_{O2}	X_{CO2}
259	2.00	0.10	0.15
260	2.00	0.10	0.25
261	2.00	0.10	0.50
262	2.00	0.10	0.70
263	2.00	0.21	0.15
264	2.00	0.21	0.25
265	2.00	0.21	0.50
266	2.00	0.21	0.70
267	3.00	0.10	0.15
268	3.00	0.10	0.25
269	3.00	0.10	0.50
270	3.00	0.10	0.70
271	3.00	0.21	0.15
272	3.00	0.21	0.25
273	3.00	0.21	0.50
274	3.00	0.21	0.70
Fuel: CH_3OH			
275	2.00	0.10	0.15
276	2.00	0.10	0.25
277	2.00	0.10	0.50
278	2.00	0.10	0.70
279	2.00	0.21	0.15
280	2.00	0.21	0.25
281	2.00	0.21	0.50
282	2.00	0.21	0.70
283	3.00	0.10	0.15
284	3.00	0.10	0.25
285	3.00	0.10	0.50
286	3.00	0.10	0.70
287	3.00	0.21	0.15
288	3.00	0.21	0.25
289	3.00	0.21	0.50
290	3.00	0.21	0.70